Oxygen-Phytoplankton-Zooplankton Dynamics in a Spatial-Temporal PDE Model

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TITLE: Oxygen-Phytoplankton-Zooplankton Dynamics in a Spatial-Temporal PDE Model

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Abstract

Ocean oxygen concentration has a strong effect on the entire marine environment, and factors such as climate change, eutrophication, and upwelling have created more intense swings in oxygen content leading to extreme ecological consequences. It is important to model dissolved oxygen content so that there is a better framework to predict future oxygen concentration lows, and see the warning signs. Along with the dissolved oxygen concentration, this project aims to model phytoplankton and zooplankton concentrations in order to capture the dynamics between all three variables. A spatial component is added to the model to understand the spacial-temporal scales of oxygen minimum zones.

Acknowledgements

Thank you to Dr. Choboter for the helpful advising and guidance on this project.

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1 Introduction

Dissolved oxygen plays a key role in maintaining a healthy marine environment. Oxygen concentration affects the production of phytoplankton, which affects the zooplankton population, which affects the small fish population, and so on until the entire marine (and partially terrestrial) ecosystem is connected. It is known that oxygen concentration varies across space, as illustrated in Figure 1. However, this is not a constant distribution across time. Diffusion and other natural events lead to shifts in concentration that can persist for long periods of time. More recently, due to the effects of anthropogenic climate change, ocean warming, agricultural runoff eutrophication, and other less studied factors, dissolved oxygen concentration often plummets. This creates oxygen minimum zones, as shown in Figure 2, which lead to massive fish die offs and consequently harsh ecological and economical consequences [3].

Studying the connection between oxygen content, phytoplankton, and zooplankton could give us a better understanding on how the oxygen minimum zones come to be, and the warning signs we might be able to see before a catastrophe. Various researchers have used systems of ODEs to model the dynamics of oxygen, phytoplankton, and zooplankton with respect to time. Sekerci and Petrovski, [1] and [6], developed a non-linear ODE model that they used to study the time dependence each of the concentrations of oxygen, phytoplankton, zooplankton. In [1], horizontal spatial dependence was also incorporated into the model, while in [2], vertical spatial distribution was introduced. Other researchers have studied different aspects of the problem. For example, Chowdhury et al. [7] looked in detail at a model that included two time scales and exhibited patterns for motion.

We analyzed an ODE model of oxygen, phytoplankton, and zooplankton to infer steady state solutions based on different environmental parameters. Then we implemented a spatial component to consider the horizontal distribution of our variables. We found oscillatory behavior across time and conducted a Fourier analysis on our data.

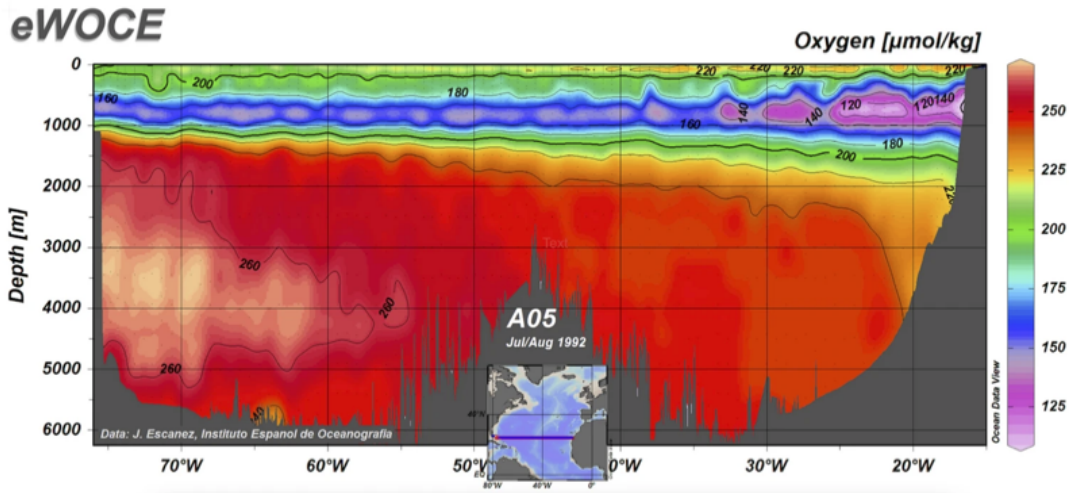


Figure 1: Dissolved oxygen profile from a transect across the Atlantic Ocean from Florida to the coast of Africa. Figure from [2].

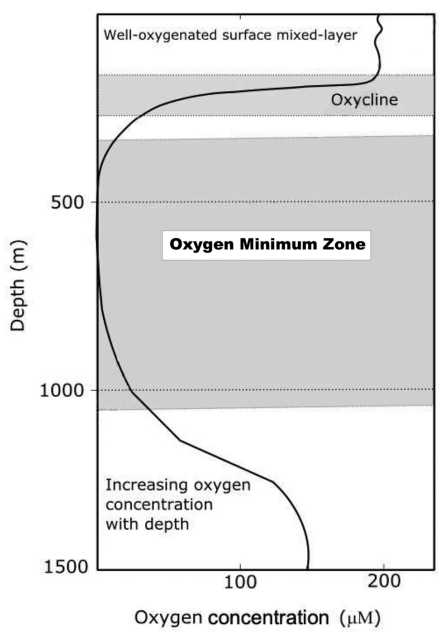


Figure 2: Typical structure of the oxygen vertical distribution in the case of an oxygen minimum zone. The black curve shows the concentration of dissolved oxygen. Figure from [2].

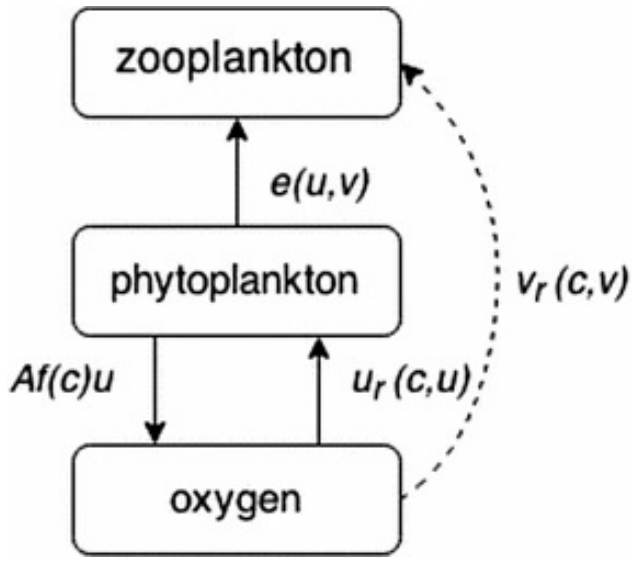


Figure 3: A conceptual figure of oxygen-phytoplankton-zooplankton relationship, from [1]. $Af(c)u$ describes phytoplankton production of oxygen, $u_r(c, u)$ and $v_r(c, u)$ describe respiration, and $e(u, v)$ describes zooplankton consumption of phytoplankton.

2 Model Construction

We began by reconstructing models created in [1] and [2]. A conceptual model including oxygen, phytoplankton, and zooplankton is created to include the minimum amount of detail for realistic results. The dynamics between the three variables is summarized by Figure 3 and equations 1-3. This first implementation did not involve a spatial component, allowing us to more easily determine the individual effects of specific parameters and determine when the solution stays steady or goes to 0. Later we added a spatial component to the model, as done in [2], to see if we could create oxygen minimum zones that appeared and disappeared across space and time.

2.1 Equations and Methods

The main equations used in the non-spatial model are below. The functions c, u, and v are the concentrations of oxygen, phytoplankton, and zooplankton, respectively, that the model hopes to

predict,

$$\frac{dc}{dt} = \frac{Au}{c+1} - \frac{\delta uc}{c+c_2} - \frac{\nu cv}{c+c_3} - c \tag{1}$$

$$\frac{du}{dt} = \left(\frac{Bc}{c+c_1} - u\right)u - \frac{uv}{u+h} - \sigma u \tag{2}$$

$$\frac{dv}{dt} = \frac{\eta c^2}{c^2+c_4^2}\frac{uv}{u+h} - \mu v. \tag{3}$$

The positive constant A describes the phytoplankton production of oxygen. δ and ν describe the oxygen consumption by phytoplankton and zooplankton respectively (respiration). The constant B accounts for phytoplankton growth through photosynthesis. The parameters c_1, c_2, and h control the saturation of the functions by limiting the where terms they are in the denominator. For example, in the term $\frac{\delta uc}{c+c_2}$, if $c < c_2$, then this term leads to exponential-like growth, but if $c > c_2$, then the growth is limited. The constants σ and μ describe the mortality rates of phytoplankton and zooplankton, and η accounts for the efficiency of the phytoplankton consumption by zooplankton.

We numerically solved these equations using the well known MATLAB function ode45 which is a differential equation solver based on an explicit Runge-Kutta (4,5) formula [4]. The results of these numerical simulations are presented in section 3.1.

The addition of the diffusion term allows for the horizontal movement of oxygen, phytoplankton, and zooplankton,

$$\frac{\partial c}{\partial t} = D_t\frac{\partial^2 c}{\partial x^2} + \frac{Au}{c+1} - \frac{\delta uc}{c+c_2} - \frac{\nu cv}{c+c_3} - c \tag{4}$$

$$\frac{\partial u}{\partial t} = D_t\frac{\partial^2 u}{\partial x^2} + \left(\frac{Bc}{c+c_1} - u\right)u - \frac{uv}{u+h} - \sigma u \tag{5}$$

$$\frac{\partial v}{\partial t} = D_t\frac{\partial^2 v}{\partial x^2} + \frac{\eta c^2}{c^2+c_4^2}\frac{uv}{u+h} - \mu v. \tag{6}$$

Diffusion-like partial differential equations are a challenge to solve numerically because they require schemes with strong stability. To find our solutions, we implemented the Crank-Nicolson method throughout numerical simulations (see Appendix A.1 for code). Crank-Nicolson is a standard, implicit, and unconditionally stable method used to solve diffusion equations [5].

We define the initial conditions for the spatial model as below. The values of $\overline{c}, \overline{u}, \overline{v}$ are the stable values of c, u, v from the non-spatial model with parameters from table 1 using initial conditions $(0.385, 0.3, 0.1)$,

$$c(x,0) = \overline{c} \tag{7}$$

$$u(x,0) = \overline{u} \tag{8}$$

$$v(x,0) = \left(x - \frac{L}{2}\right)\frac{\epsilon}{L} + \overline{v}, \tag{9}$$

$$(\overline{c}, \overline{u}, \overline{v}) = (0.4584, 0.464, 0.0836),$$

and therefore

$$c(x,0) = 0.4584 \tag{10}$$

$$u(x,0) = 0.464 \tag{11}$$

$$v(x,0) = \left(x - \frac{1000}{2}\right)\frac{0.02}{1000} + 0.0836. \tag{12}$$

2.2 Parameters

We began numerical simulations for the non-spatial model using the same parameter values as [2] and summarized in Table 1. To conduct a stability analysis of steady state solutions, we ran the code with varying values of A and c_1 to see if we get the same results as in [2]. After conducting 300 iterations in the same manner as the paper, we found that the overall distribution of the parameter plane looked similar, but the specific values where the solution went from stable to unstable differed.

Figure 4 shows how varying the values of the parameter A (phytoplankton production of oxygen) and c_1 (the half saturation constant limiting phytoplankton growth) affected the steady state solutions. Red points represent those values where the solution goes to 0 for that pair of values and all values where A is smaller for the same c_1. Green points represent those values where the

Parameters	
Name	Value/Range
A	[0,4]
δ	1
c_1	0.7
c_2	1
c_3	1
c_4	1
ν	0.01
B	1.8
σ	0.1
η	0.7
μ	0.1
h	0.1

Table 1: Parameter values used throughout numerical simulations.

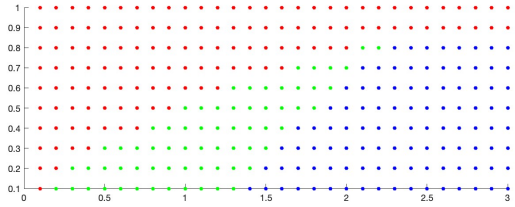
solution persists stably and non-trivially. Blue points represent those values where the solution goes to 0 for that pair of values, but for a smaller A at the same value of c_1, the solution had been non-zero.

We can directly compare our finding from Figure 4a to those from [2] in Figure 4c. Our red points correspond to those in domain 1, green points to those in domains 2, 3, and 4, and blue points to those in domain 5. The general area of the transitions is the same, but specific values differ. Upon a thorough check of our code we have not found in error on our part yet, and due to the lack of description of the specific algorithms used in previous papers, we are unable to conclude how we came upon different results.

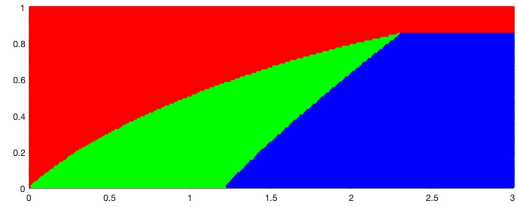
3 Model Outcomes

3.1 Time-Dependent Model

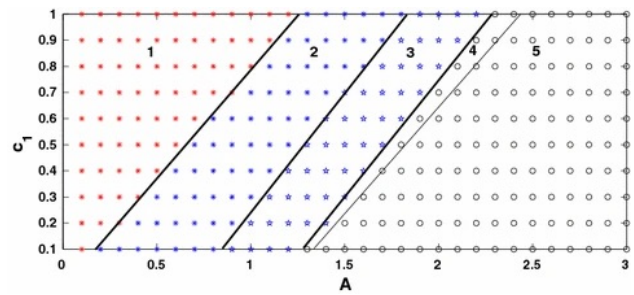
We began by running the time-dependent (non-spatial) ODE model in MATLAB, making use of the ode45 function. Through varying the values of A, representing phytoplankton production of oxygen, we were able to see what different steady states looked like. Results can be viewed both as concentrations over time in Figure 5 and as trajectories in 3D phase space in Figure 6. At values



(a) 300 iterations



(b) 30000 iterations



(c) Parameter plane of (A, c_1) from [2]

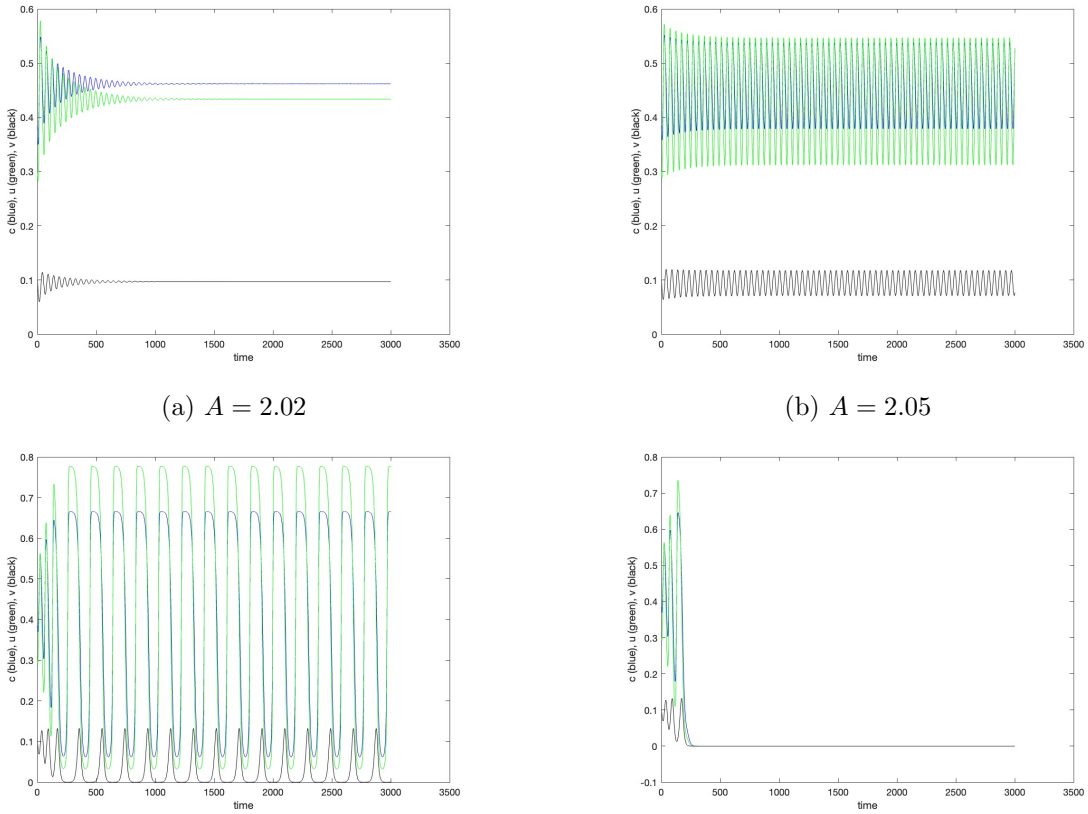
Figure 4: Parameter planes of A on the x-axis and c_1 on the y-axis with different amounts of iterations. For a and b, red and blue points represent solutions that went to 0, and green points represent the solutions that persisted stably.

of A at or slightly below 2.02 (see Figure 4b with $c_1 = 0.7$ and points in the green domain), the solution exhibited damped oscillations that settled on values near the initial values. When the value of A was so that it was on the border of the green domain in Figure 4b, solutions would be highly oscillatory and values would continue varying throughout time. As A grew larger than these critical values, solutions would fall down to 0.

We can also observe that oxygen and phytoplankton concentrations are almost in phase with phytoplankton slightly leading oxygen concentrations (Figure 7). This aligns with scientific knowledge surrounding seasonal phytoplankton blooms (also described in [3]) and confirms that the model is capturing natural phenomena with minimal components.

3.2 Spatial

Implementing the spatial model required the addition of more elements such as the new initial values graphed in Figure 8. We found that the values of A that created the three distinct solution forms in the non-spatial case (solutions from Figure 6 and the values of A described in Figure 4b),



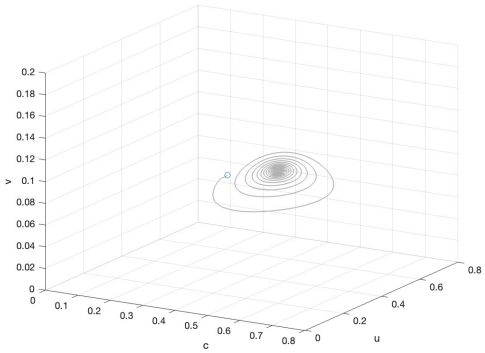
(a) $A = 2.02$

(b) $A = 2.05$

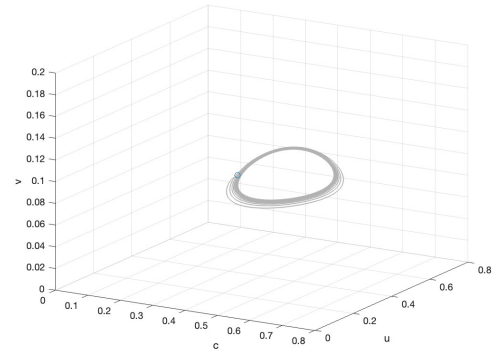
(c) $A = 2.0927$

(d) $A = 2.093$

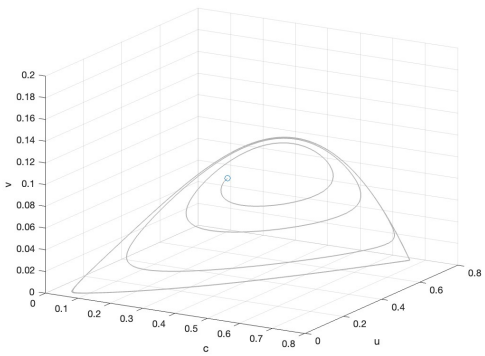
Figure 5: Plots of solutions over time for various values of A. The blue line is oxygen concentration (c), the green line is phytoplankton concentration (u), and the black line is zooplankton concentration (v)



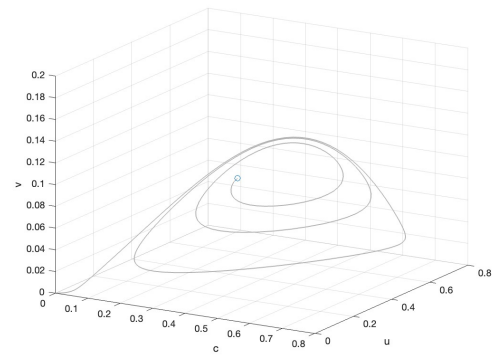
(a) $A = 2.02$



(b) $A = 2.05$



(c) $A = 2.0927$



(d) $A = 2.093$

Figure 6: Trajectories of the oxygen-phytoplankton-zooplankton (c, u, v) dynamical system shown in the corresponding 3D phase space with varying values of A. The initial points are set at $(0.385, 0.3, 0.1)$

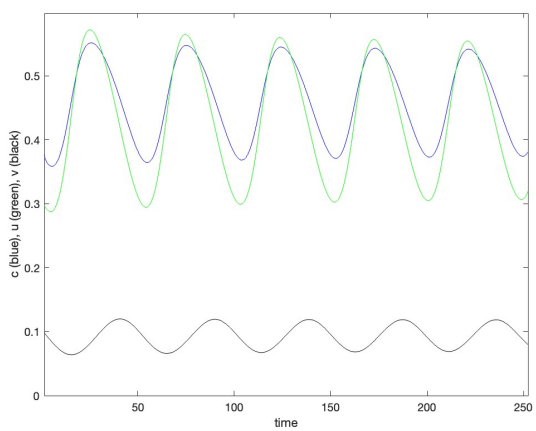


Figure 7: Plot of solutions from $t = 0$ to $t = 250$

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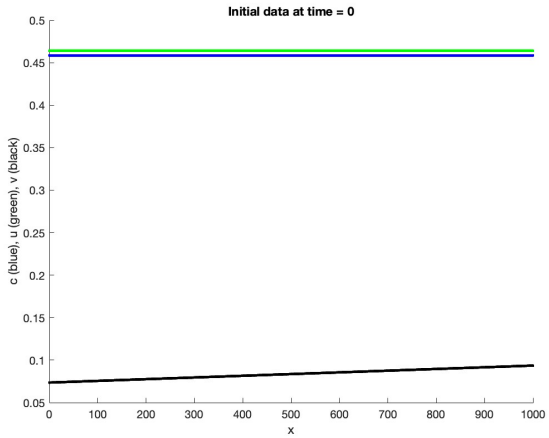


Figure 8: Initial conditions using equations 7-9.

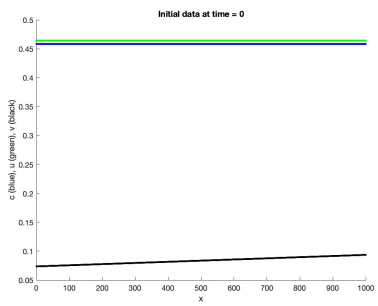
must be lowered slightly to create three distinct solution forms for the spatial model.

At lower values of A that created stable solutions ($A = 1.9$ for instance), the concentrations quickly diffused equally across space, as summarized by Figure 9.

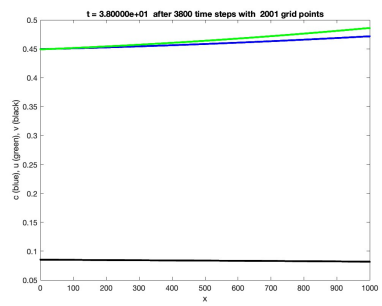
At the values of A where the solution borders between stable and disappearing to 0, the concentrations exhibit highly oscillatory behavior that shows no clear pattern, as shown in Figure 10. At the start, for a significant amount of time, all concentrations drop to 0 across space except at two spatial points. Due to the effects of diffusion, concentrations level back out over time. For some time there appears to be little spatial effects for the middle area, for all concentrations. As time continues though, more irregular oscillations occur at all points in space. When concentrations drop to zero for small sections of space, the diffusion effects bring the concentrations back up.

In the case of $A = 2.1$ we can also investigate how the first long drop in concentrations is formed. From Figure 11 we can see that the drop in oxygen preceded the drop in phytoplankton concentrations, and the recovery of the phytoplankton occurred sooner. This connects to how, following a phytoplankton bloom and subsequent large phytoplankton die off and decay, dissolved oxygen content drops.

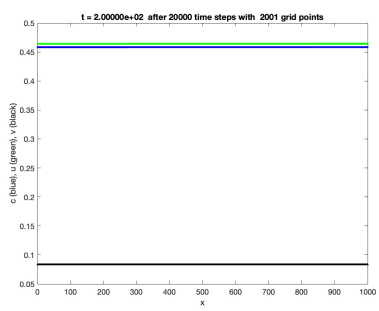
At sufficiently large values of A, the solution disappears to 0, as summarized by Figure 12.



(a) $A = 1.9$, at time $t = 0$.

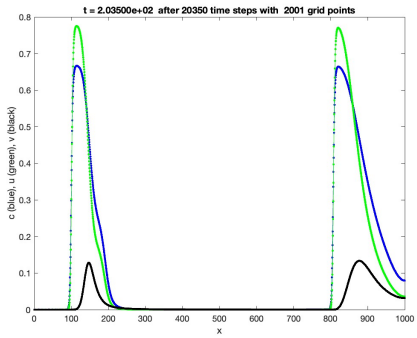


(b) $A = 1.9$, at time $t = 38$ from start.

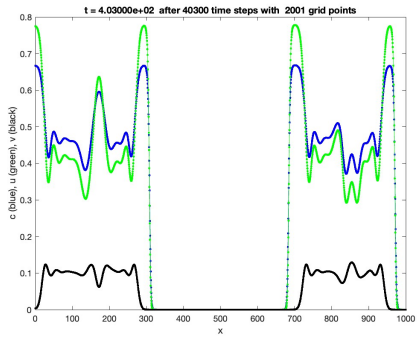


(c) $A = 1.9$, after $t = 200$. Stable solution is reached, constant throughout space.

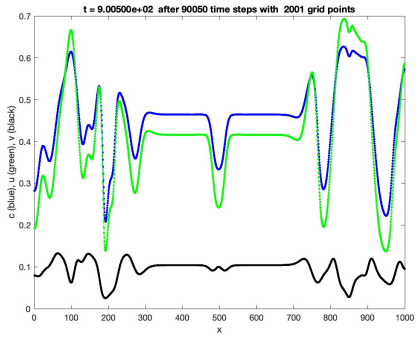
Figure 9: Snapshot graphs of a stable solution with c in blue, u in green, and v in black.



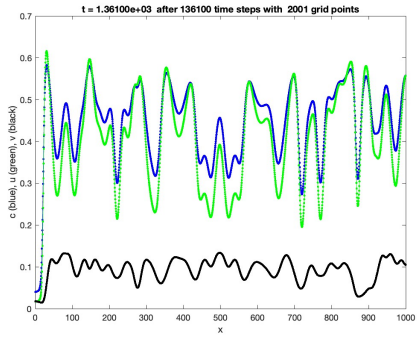
(a) After 20350 time steps.



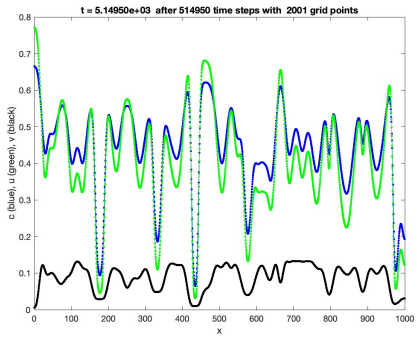
(b) After 40300 time steps.



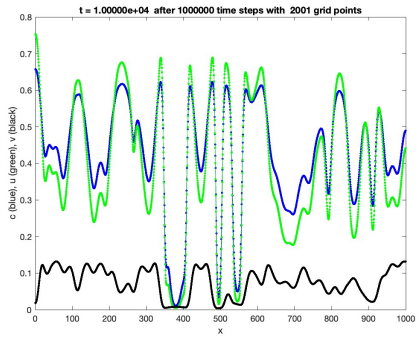
(c) After 90050 time steps.



(d) After 136100 time steps.



(e) After 514950 time steps.



(f) After 1000000 time steps.

Figure 10: Snapshot graphs of oxygen (blue), phytoplankton (green), and zooplankton (black) using $A = 2.1$ and initial conditions from equations 10-12.

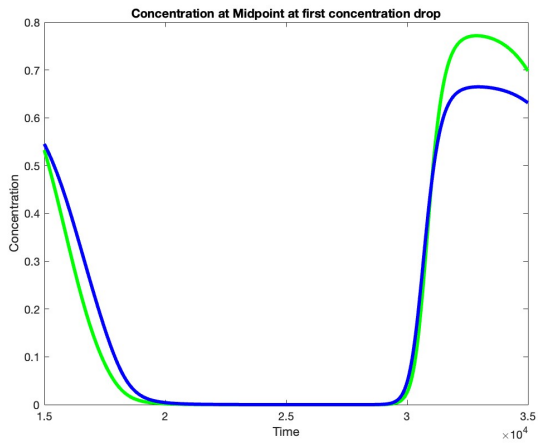


Figure 11: Oxygen (blue) and phytoplankton (green) concentrations between 15000 and 35000 time steps at $x = 500$. Zooplankton is not shown because the values are small on this scale.

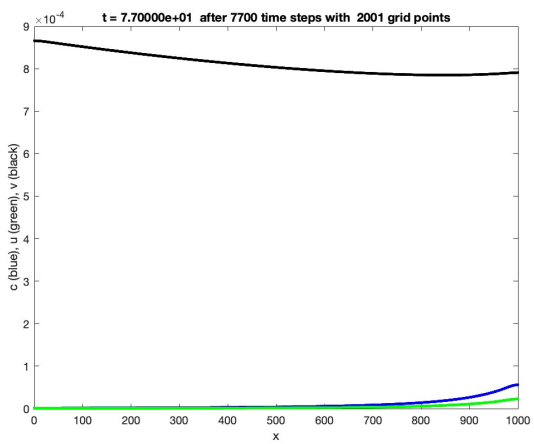


Figure 12: $A = 2.2$, after 7700 time steps. Note that the y-axis labels range from 0 to .0009. For all greater time values, zooplankton (the black line) will continue to decrease slowly.

3.3 Comparisons

Despite the parameter planes suggesting different behaviors of the models created by [2] and us, solutions to both the spatial and non-spatial cases exhibited the same behavior. In the non-spatial case it is much easier to compare solutions, since we can clearly see and calculate what values the concentrations land on in the long run. The spatial models are much harder to compare. [2] does not mention the lower and higher ranges of A that made the solution either become constant across space or disappear altogether. We can qualitatively look at our solutions that exhibited highly oscillatory behavior at the border values of A and compare them to the results from previous work. In both cases oxygen minimum zones would appear across time at different places in space, but they would return to non-zero values as the oxygen from nearby areas diffused in.

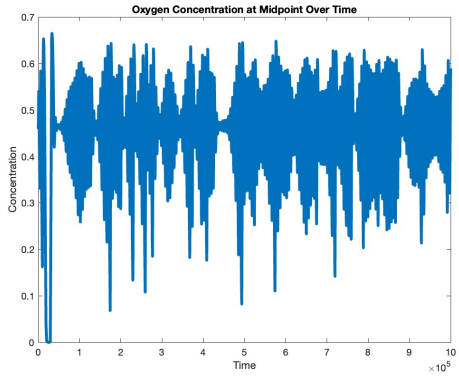
3.4 FFT Analysis

We were curious to learn more about the oscillatory behavior of solutions and investigate how oxygen minimum zones appear and disappear. The highly complex solutions can be decomposed into simpler and more understandable components. Figure 13 shows the irregular patterns of concentrations we want to understand.

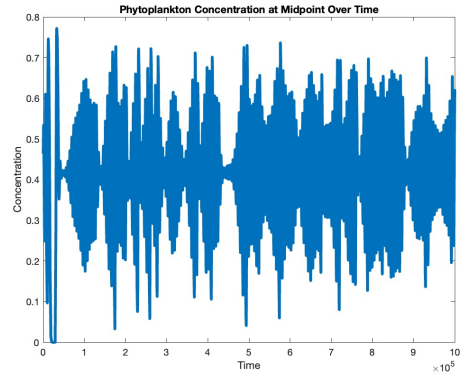
We studied the Fourier transforms of given concentrations at points in space across time. An example of the code we used to create Figure 14 is in Appendix A.2. From the single sided fft amplitude spectrums we found that there is one major peak among all the locations that we tried, both for oxygen and phytoplankton. The peak is also at around the same value of about 0.19. This tells us that every oscillation has a period of about 5.26 time units.

4 Discussion

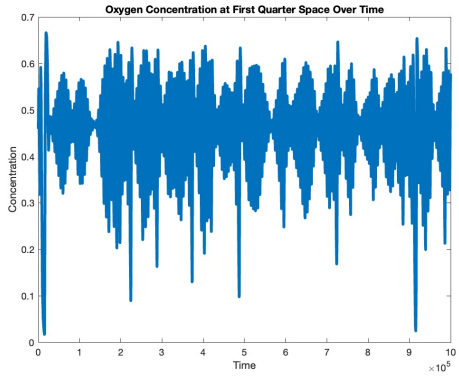
In this project we were able to recreate a non-spatial and spatial model of oxygen-phytoplankton-zooplankton dynamics developed in [1] and [2]. We implemented our own code in MATLAB and found solutions similar to those in the papers. The exact values of solutions varied from the papers, but the types of solutions (constant, oscillatory, zero) we uncovered were the same. Along with



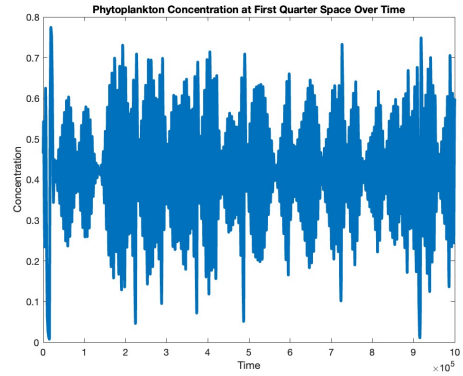
(a) Oxygen concentration at $x = 500$ over time.



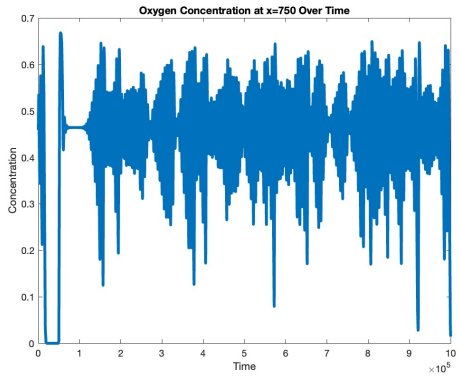
(b) Phytoplankton concentration at $x = 500$ over time.



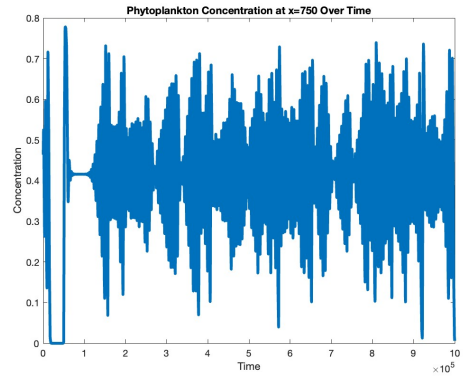
(c) Oxygen concentration at $x = 250$ over time.



(d) Phytoplankton concentration at $x = 250$ over time.



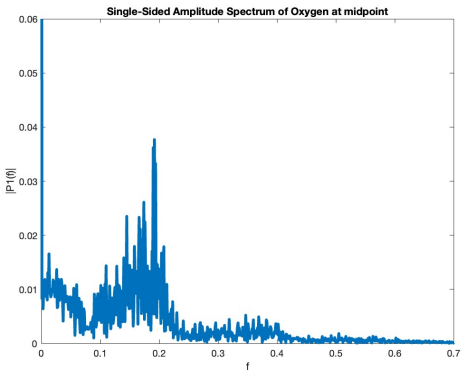
(e) Oxygen concentration at $x = 750$ over time.



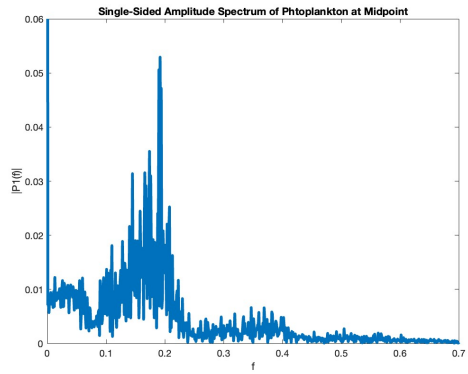
(f) Phytoplankton concentration at $x = 750$ over time.

Figure 13: Concentrations of oxygen and phytoplankton across time at varying locations in space.

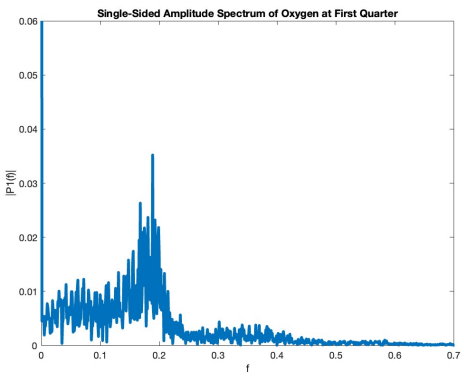
Figure 14: Zoomed in single sided fft amplitude spectrum for oxygen (a), (c), (e), and phytoplankton (b), (d), (f).



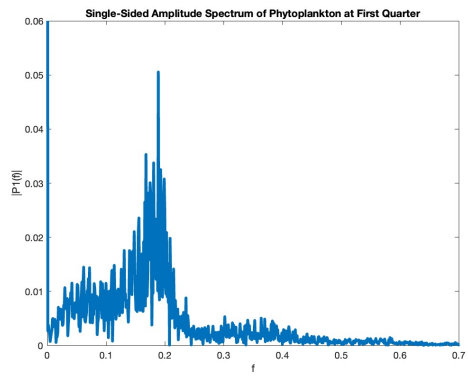
(a) Set at $x = 500$



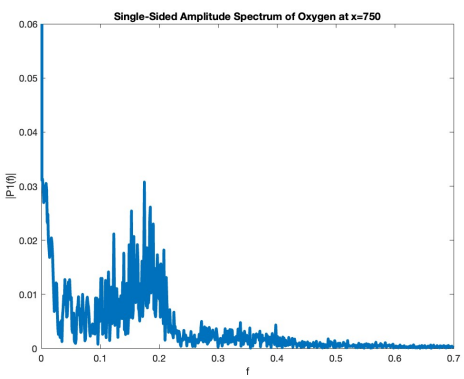
(b) Set at $x = 500$



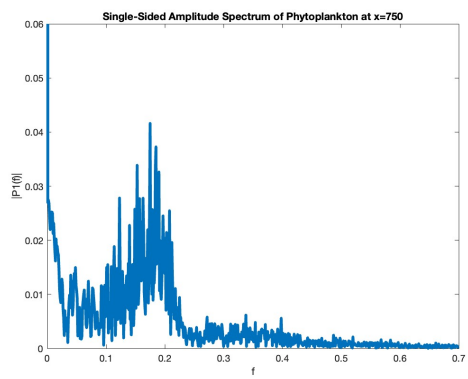
(c) Set at $x = 250$



(d) Set at $x = 250$



(e) Set at $x = 750$



(f) Set at $x = 750$

analyzing the parameters and their affects on the solutions, we conducted a Fourier analysis on solutions to the spatial model to discover that a common spatial scale is shared among the three concentrations across space and time.

Our solutions tell us that sometimes oxygen concentrations decline to zero but are able to recover when some of the model parameters are in a certain range (that is when A, the phytoplankton production of oxygen, and c_1, the half saturation constant limiting phytoplankton growth, lie in the ranges described in Figure 4). Our results also showed us that oxygen minimum zones have spatial dependencies described in Figure 10. This aligns with known findings and data, and the temporal consistencies we found doing the Fourier analysis tell us that a common time scale is present across space for oxygen, phytoplankton, and zooplankton concentrations.

We were not able to find a reason for the disparities found among our work and [2] regarding parameters (see Figure 4). This could be due to different techniques used to solve the equations, and more information is needed.

Future work on this project could involve adding more of the dynamics of environmental forces, such as upwelling or temperature, which are known to have effects on oxygen distribution in the ocean as well. It would also be interesting to do more analysis of the model to see if there are ways to predict when there will be a drop in oxygen concentration at a certain point in space, since the oscillatory solutions to the model simulated many sudden drops. Along with the phytoplankton oxygen production A and the saturation parameter c_1 we looked at, future work could analyze different parameters such as B, the phytoplankton growth through photosynthesis.

A Matlab code

A.1 Crank-Nicolson Implementation

```
clf              % clear graphics
hold on          % Put all plots on the same graph (comment out if desired)

ax = 0;
bx = 1000; % this is L from the paper if ax=0.
kappa = 1;               % heat conduction coefficient:
tfinal = 10000;                % final time

m = 2000; % set to 2L for delta x = 0.5.
h = (bx-ax)/(m);         % h = delta x
x = linspace(ax,bx,m+1)';  % note x(1)=ax and x(m+2)=bx
                           % u(1)=g0 and u(m+2)=g1 are known from BC's
%k = 4*h;        % time step
k = 0.01;         % time step

nsteps = round(tfinal / k);    % number of time steps
nplot = 50;      % plot solution every nplot time steps
                 % (set nplot=2 to plot every 2 time steps, etc.)
%nplot = nsteps;  % only plot at final time

if abs(k*nsteps - tfinal) > 1e-5
   % The last step won't go exactly to tfinal.
   disp(' ')
   disp(sprintf('WARNING *** k does not divide tfinal, k = %9.5e',k))
   disp(' ')
end

% initial condition function:
% For Gaussian initial conditions u(x,0) = exp(-beta * (x-0.4)^2)
beta = 0.00001;
uinit = @(x,t) exp(-(x-500).^2 / (4*kappa*t + 1/beta)) / sqrt(4*beta*kappa*t+1);
% initial conditions:
c0 = .4584*ones(2001,1);
u0 = .464*ones(2001,1);
v0eqn = @(x) (x-bx/2)*(.02/bx)+.0836;
v0 = v0eqn(x)

% Each time step we solve MOL system U' = AU + g using the Trapezoidal method

% set up matrices:
r = (1/2) * kappa* k/(h^2);
e = ones(m+1,1);
A = spdiags([e -2*e e], [-1 0 1], m+1, m+1);
% this is for the no-derivative BCs:
A(1,2) = 2; A(m+1,m) = 2;
A1 = speye(m+1) - r * A;
A2 = speye(m+1) + r * A;
```

```
% initialize u and plot:
tn = 0;
c = c0;
u = u0;
v = v0;

plot(x,c,'b.-')
plot(x,u,'r.-')
plot(x,v,'g.-')
title('Initial data at time = 0')

input('Hit <return> to continue  ');

% main time-stepping loop:
for n = 1:nsteps
     tnp = tn + k;   % = t_{n+1}
     % boundary values u(0,t) and u(1,t) at times tn and tnp:
     g0n = 0;
     g1n = 0;
     g0np = 0;
     g1np = 0;

     % compute right hand side for linear system:
     uint = u(1:(m+1));   % interior points (unknowns)
     cint = c(1:(m+1));
     vint = v(1:(m+1));
     % p = vector of parameters = (A,delta,c1,c2,c3,c4,nu,B,sigma,eta,mu,h)
     p=[2.1 1 .7 1 1 1 .01 1.8 .1 .7 .1 .1];
     rhsc = A2*cint + k*(((p(1)*uint)./(cint+1))-((p(2)*uint.*cint)./(cint+p(4)))-((p(7)*cint.*vint)./(cint+p(5)))-cint);
     rhsu = A2*uint + k*(((p(8)*cint./(cint+p(3))-uint).*uint)-((uint.*vint)./(uint+p(12)))-p(9)*uint);
     rhsv = A2*vint + k*((((p(10)*(cint.*cint))./((cint.*cint)+(p(6)^2))).*((uint.*vint)./(uint+p(12))))-p(11)*vint);
     % fix-up right hand side using BC's (i.e. add vector g to A2*uint)
     % (leftover from Dirichlet BCs)
     %rhs(1) = rhs(1) + r*(g0n + g0np);
     %rhs(m+1) = rhs(m+1) + r*(g1n + g1np);

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
     % solve linear system:
     uint = A1\rhsu;
     vint = A1\rhsv;
     cint = A1\rhsc;
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

     % augment with boundary values for plots:
     % (no-flux BC is zero-derivative BC)
     % (leftover from Dirichlet BCs)
     u = [uint];
     v = [vint];
     c = [cint];

     % plot results at desired times:
     if mod(n,nplot)==0 | n==nsteps
        plot(x,c,'b.-')
        hold on;
```

```
        plot(x,u,'r.-')
        plot(x,v,'g.-')
        drawnow;
        title(sprintf('t = %9.5e  after %4i time steps with %5i grid points',...
                      tnp,n,m+1))
        %if n<nsteps, input('Hit <return> to continue  '); end;
        if n<nsteps, pause(0.02); end;
     end

     tn = tnp;   % for next time step
     hold off;
end
```

A.2 FFT Analysis

```
    % Using FFT or one point in space across time
Y_mid_space = fft(X_mid_space);
plot(1:nsteps,X_mid_space, "LineWidth",3)
title("Oxygen Concentration at Midpoint Over Time")
xlabel("Time")
ylabel("Concentration")
%%
plot(1:nsteps,abs(Y_mid_space),"LineWidth",3)
title("Complex Magnitude of fft Spectrum")
xlabel("f")
ylabel("|fft(X)|")
%%
plot(1:nsteps,abs(fftshift(Y_mid_space)),"LineWidth",3)
title("Oxygen fft Spectrum in the Positive and Negative Frequencies")
xlabel("f")
ylabel("|fft(X)|")
P2 = abs(Y_mid_space/nsteps);
P1 = P2(1:nsteps/2+1);
P1(2:end-1) = 2*P1(2:end-1);
f = Fs/nsteps*(0:(nsteps/2));
plot(f,P1,"LineWidth",3)
xlim([0 .7])
ylim([0 .06])
title("Single-Sided Amplitude Spectrum of Oxygen at midpoint")
xlabel("f")
ylabel("|P1(f)|")
```

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